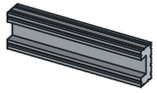
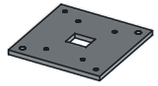
Gantry · parts

Count everything first. Quantities cover the complete gantry.



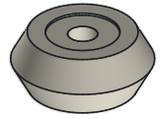
C-Beam 1500 silver LR-C-S

2×



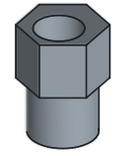
XL gantry plate 125 × 125 × 6

2×



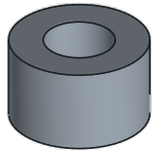
Solid V wheel OD 23.9

8×



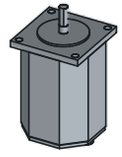
Eccentric spacer, 8 mm hex

4×



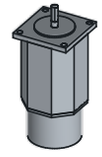
Aluminium spacer 6 mm

4×



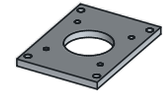
NEMA 23 · X axis

1×



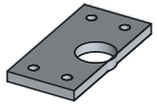
NEMA 23 + 24 V brake · Z

1×



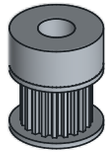
NEMA 23 motor plate

2×



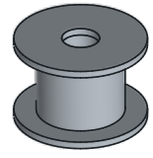
C-Beam end plate · mast top

1× not in BOM Rev A



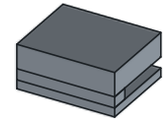
GT2 pulley 20T, 6.35 bore

1×



Idler, 9 mm belt

1×

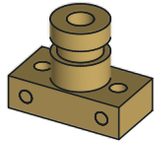


Belt clamp block

2×

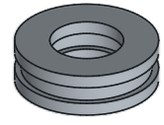
Lead screw Tr8×8, 1500

1×



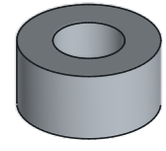
Anti-backlash nut block

1×



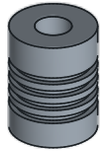
Thrust bearing, 8 mm

2×



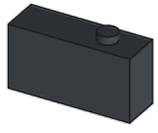
Lock collar, 8 mm

2× not in BOM Rev A



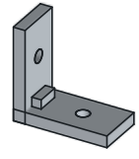
Coupler 6.35 → 8

1×



Limit switch, NC

4×



Corner bracket

4×



Arm plate 160 sq × 6.35

1×

Also: GT2 belt 9 mm (2.8 m for the loop), wiring, fuse block, E-stop and the safety relay. In the BOM, not drawn.

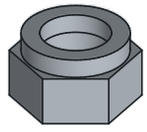
Gantry · fasteners

Every screw in the build. Lay them out in labelled pots first.



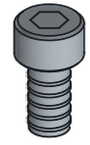
M5 × 30 low profile · wheels

8×



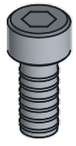
M5 nyloc · wheels, idler

9×



M5 × 10 · motor, plate, clamps

12×



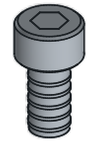
M5 × 12 · end plates, nut, arm plate

14×



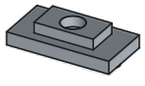
M5 × 40 socket cap · idler axle

1×



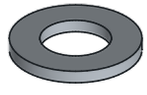
M5 × 10 · corner brackets

8×



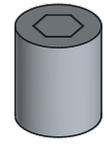
M5 T-nut, drop in

12×



M5 washer · idler

2×



M5 × 6 grub · pulley

2×



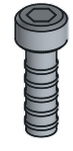
M4 × 6 grub · coupler

4×



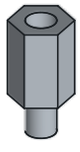
M4 × 10 socket cap · arm base

4×



M3 × 10 + T-nut · limit switches

8×



M3 standoff + M3 × 10

4×

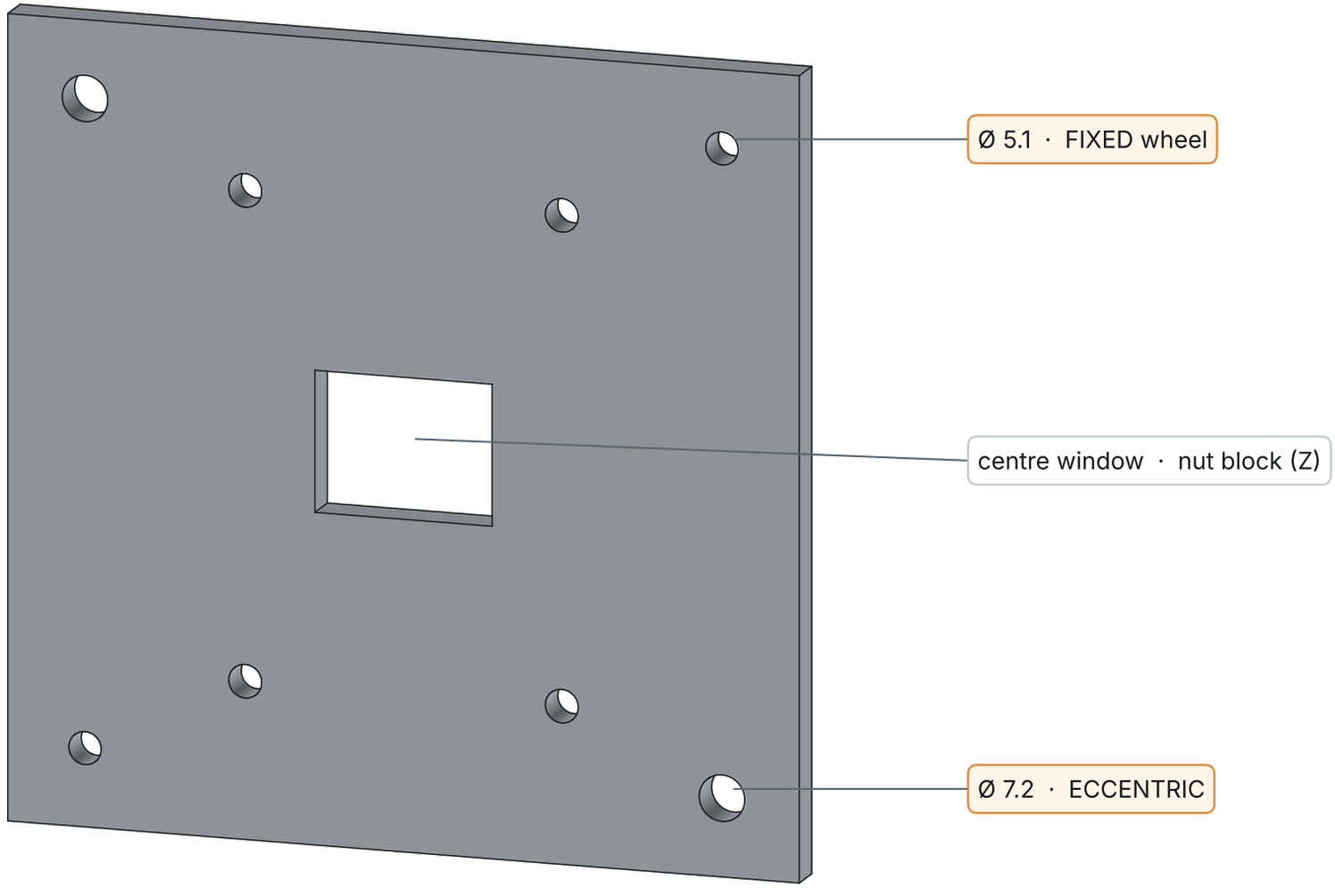
2 / 2.5 / 3 / 4 mm hex

8 mm spanner, thin

The wheel screws are M5 × 30, not the 27 of BOM Rev A: plate 6 + spacer 6 + wheel 10.23 + nyloc 5 is 27.2, so a 27 never reaches the nyloc's nylon ring.

1 Know your plate

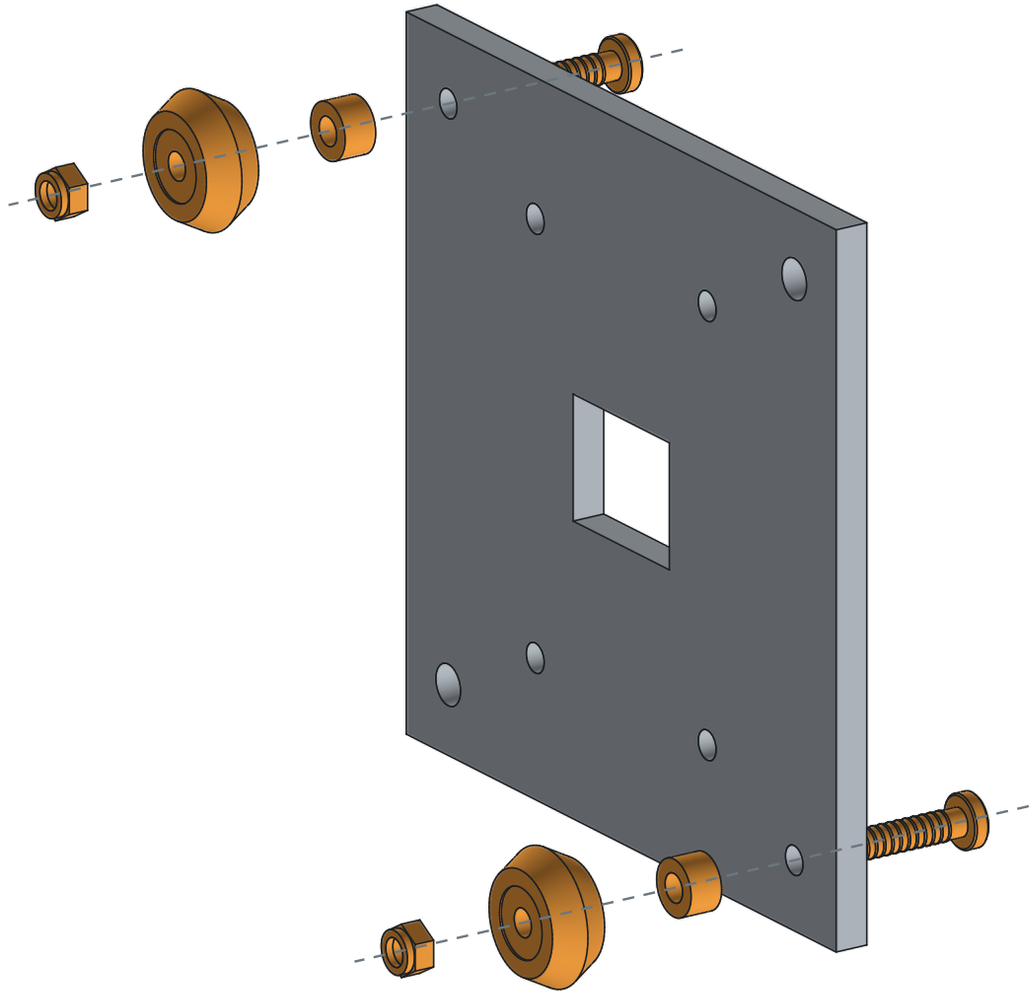
Two hole sizes. Getting them the wrong way round costs the carriage.



The two small holes sit on one diagonal and take the fixed wheels. The two large ones take the eccentric spacers, which is how you will preload the wheels against the beam. The wheels end up OUTSIDE the beam, straddling its 80 mm height, 100.6 mm apart.

2 Fixed wheels · the stack

Screw, plate, spacer, wheel, nyloc. Diagonal pair only.



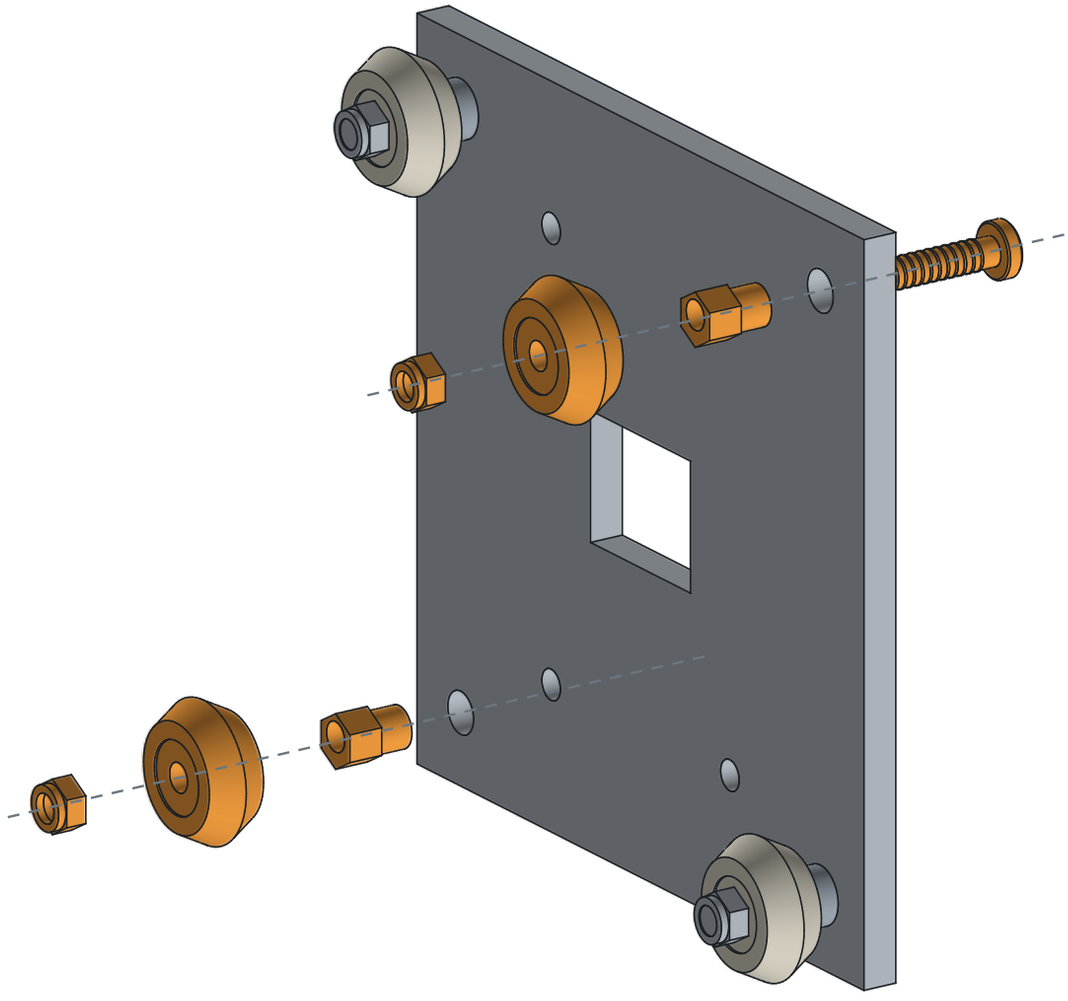
- 2× M5 × 30 low profile
- 2× Aluminium spacer 6 mm
- 2× Solid V wheel
- 2× M5 nyloc

3 mm hex · 8 mm

Bottom-left and top-right only. Snug the nylocs, do not torque them: you will re-check every one after preload, because tightening a wheel usually adds preload.

3 Eccentric wheels

The other diagonal. The hex sits between the plate and the wheel.

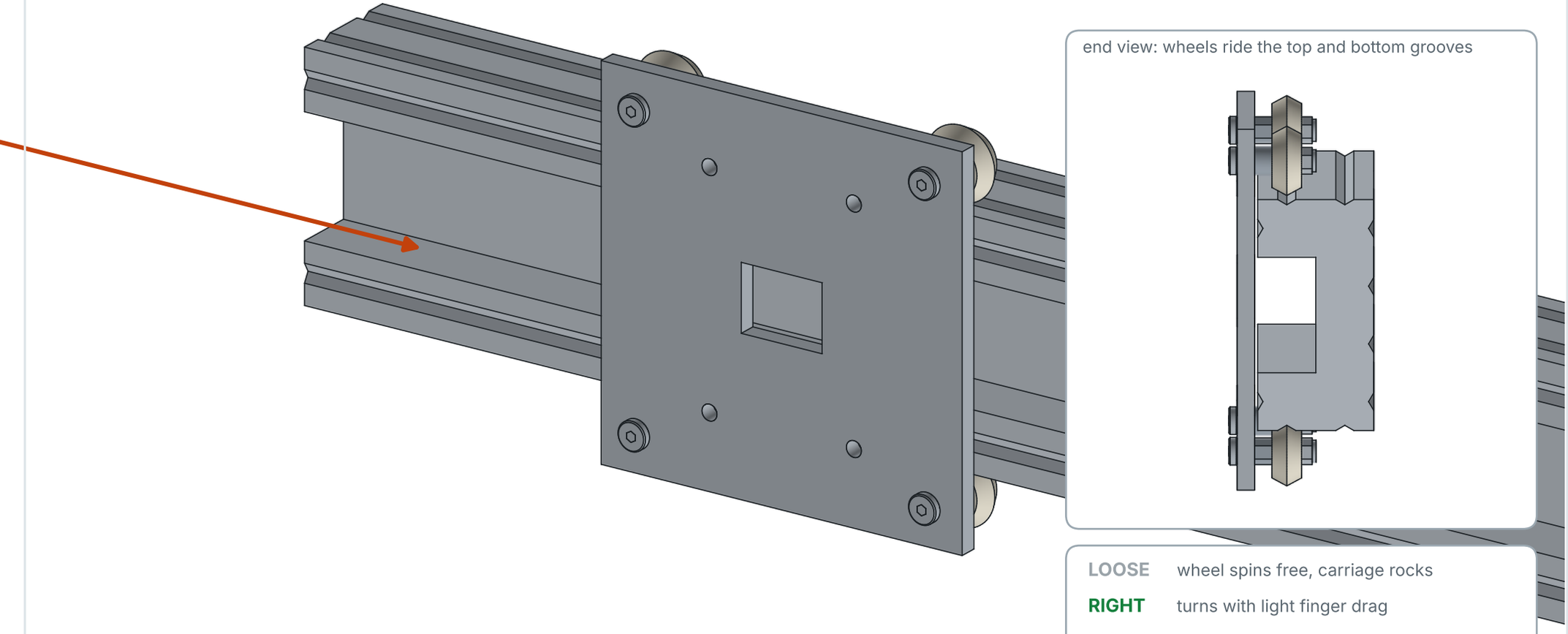


- 2× Eccentric spacer, 8 mm hex
- 2× Solid V wheel
- 2× M5 × 30 low profile
- 2× M5 nyloc

The spacer's round boss drops into the 7.2 mm hole; its hex is the spacer. Set both at their LOOSEST (wheel furthest from the beam) before fitting, or the carriage will not go on. You turn them later with a thin 8 mm spanner, from the side, between plate and wheel.

4 Onto the rail, then set the preload

Slide on from the end. Then the one adjustment that decides the machine.



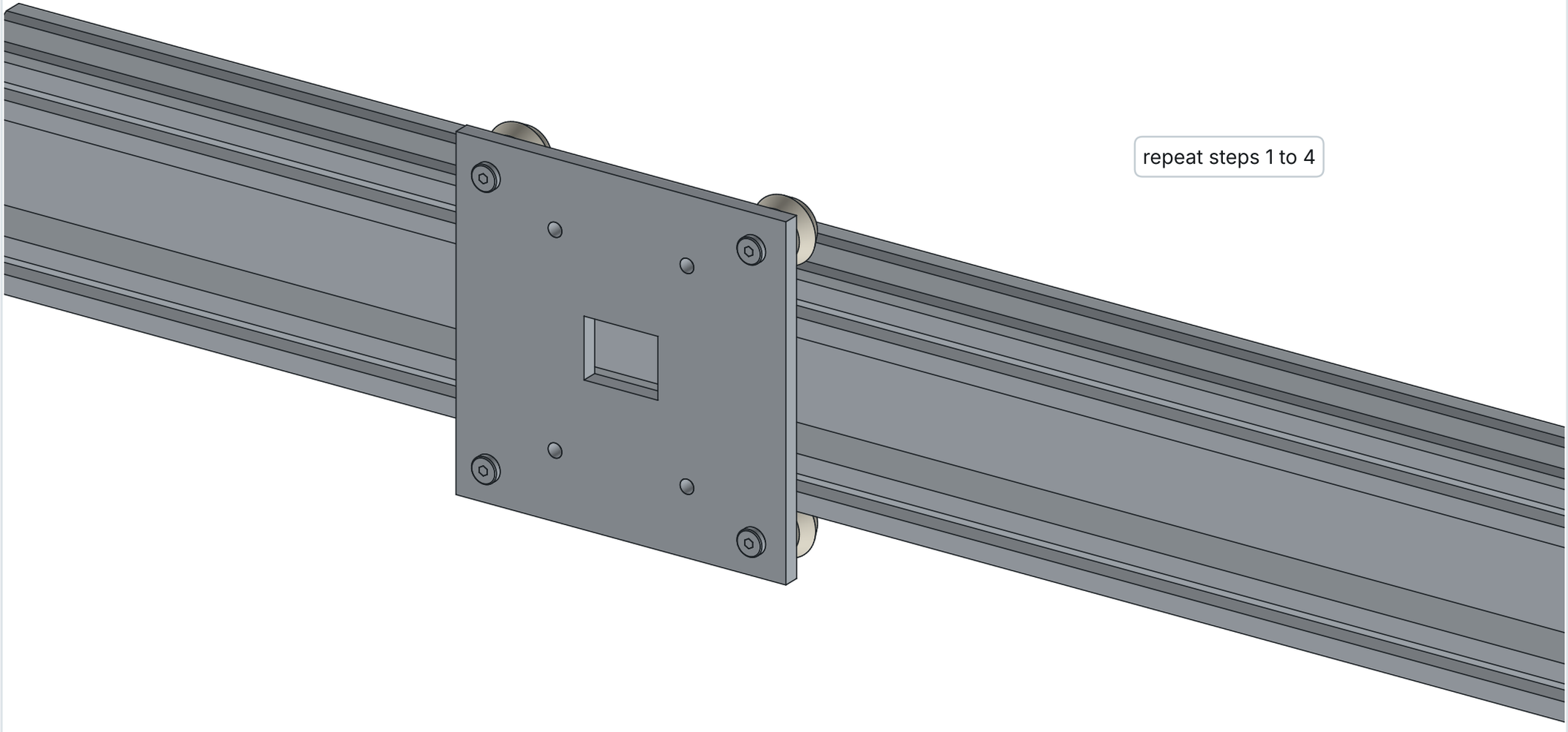
end view: wheels ride the top and bottom grooves

LOOSE	wheel spins free, carriage rocks
RIGHT	turns with light finger drag
TIGHT	hard to turn, wears flats in the wheel

Turn each eccentric until its wheel just grips the groove, then back off until it turns with light finger drag. Tighten the nyloc and re-check. Run the carriage the full length by hand: even resistance, no rock, no tight spot. Too loose is slop, and slop defeats hand-eye calibration: the camera says where the leaf is and the arm arrives somewhere else.

5 Build the second carriage

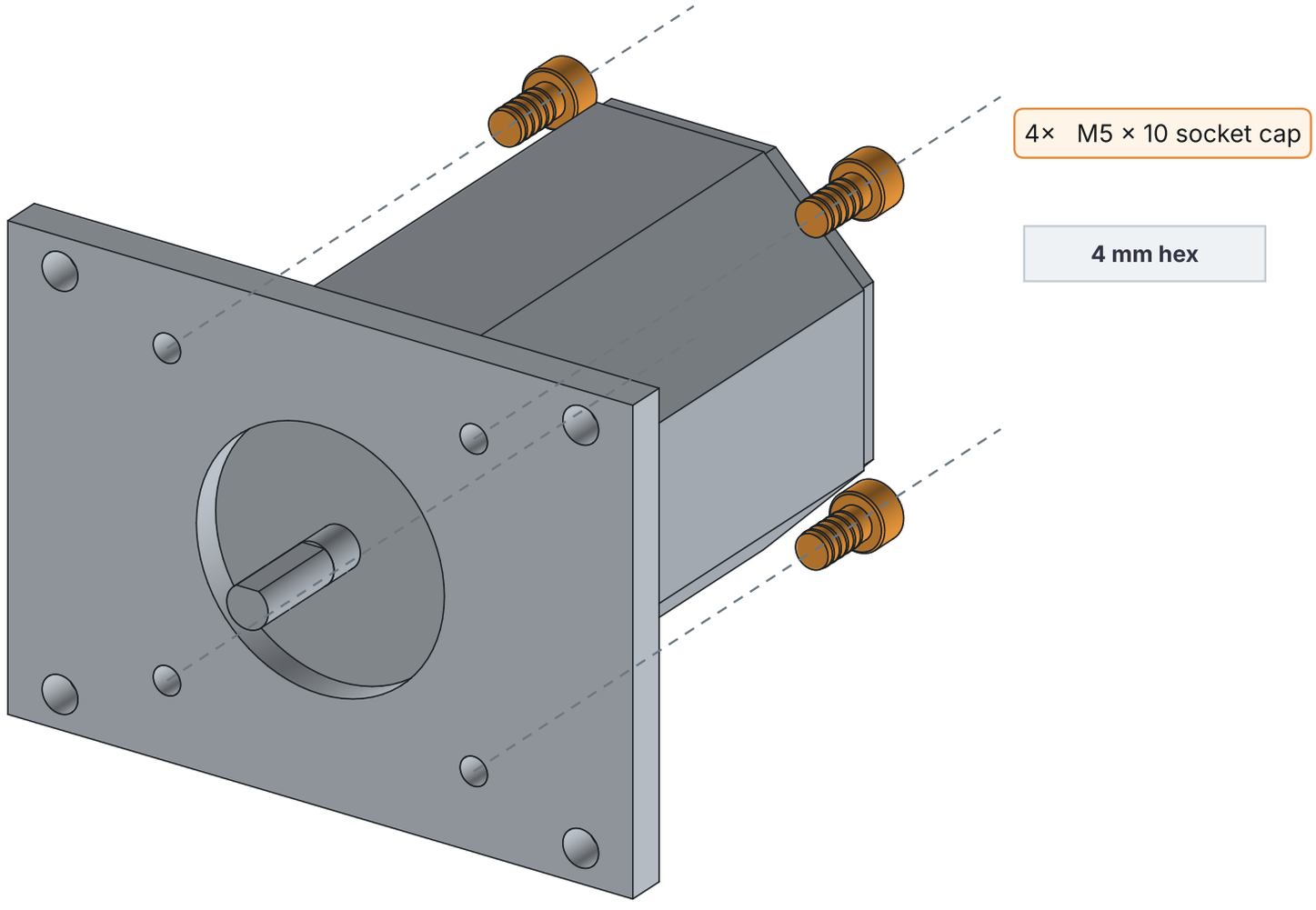
Identical. It rides the 1500 mm Z mast.



Build and preload it now, while the method is fresh. The only difference comes at step 11, when the nut block goes on its back.

6 X motor onto its plate

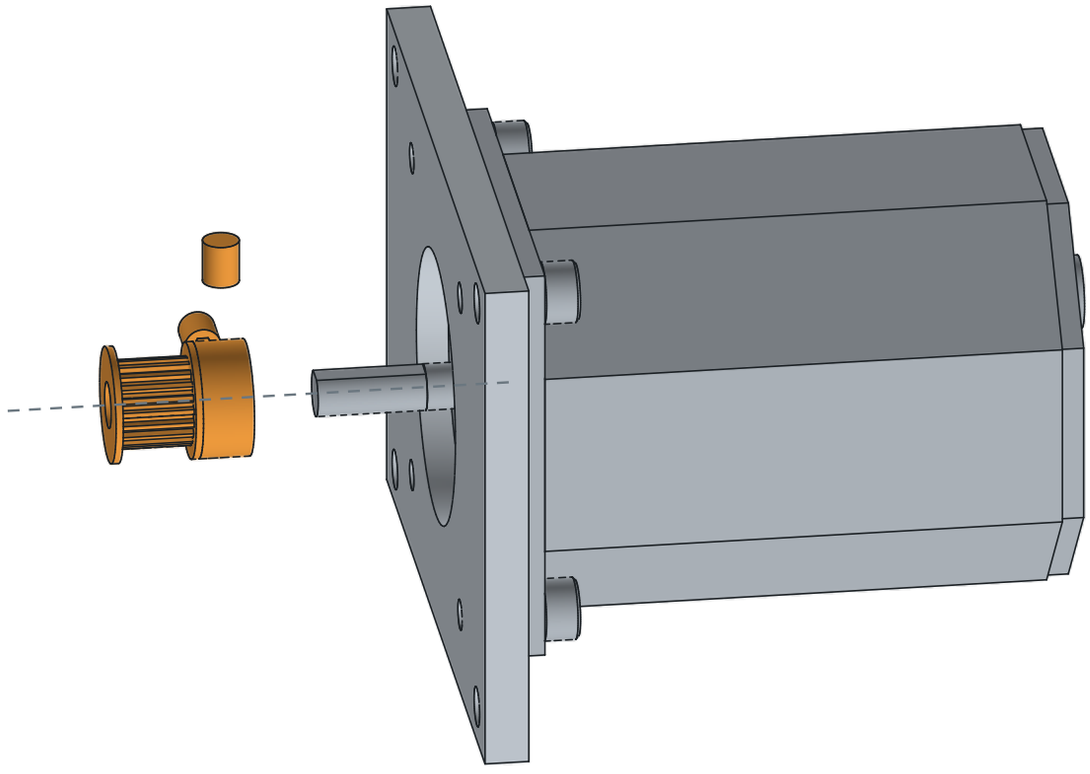
Four M5 × 10, from the MOTOR side, into the plate.



NEMA 23 flanges are drilled through, so the heads seat on the BACK of the motor flange, in the pockets beside the body, and the thread takes in the plate. If the plate's four motor holes are plain, tap them M5 first. The boss locates in the 38.1 bore: if it will not seat, something is between the faces.

7 Pulley onto the shaft

Now, while you can reach the grub screws. After step 8 you cannot.



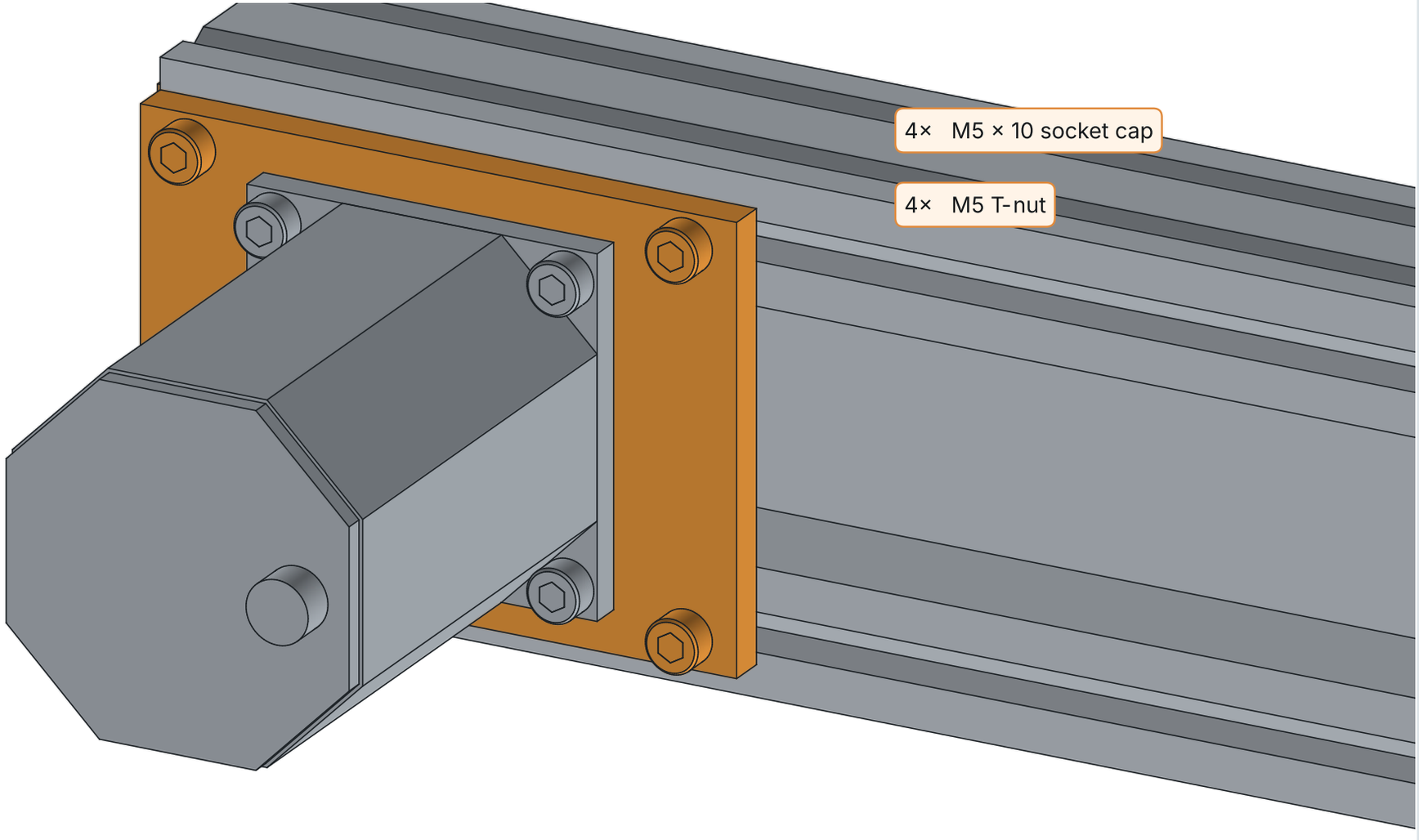
- 1× GT2 pulley 20T, 6.35 bore
- 2× M5 × 6 grub screw

2.5 mm hex

Boss toward the motor, flanges outboard: the toothed hub must finish 22 to 32 mm from the plate's back face, which is where the belt runs inside the rail. Seat the FIRST grub on the shaft flat. A grub on the round creeps under load and loses steps exactly like a current problem would.

8 Motor onto the rail's left end

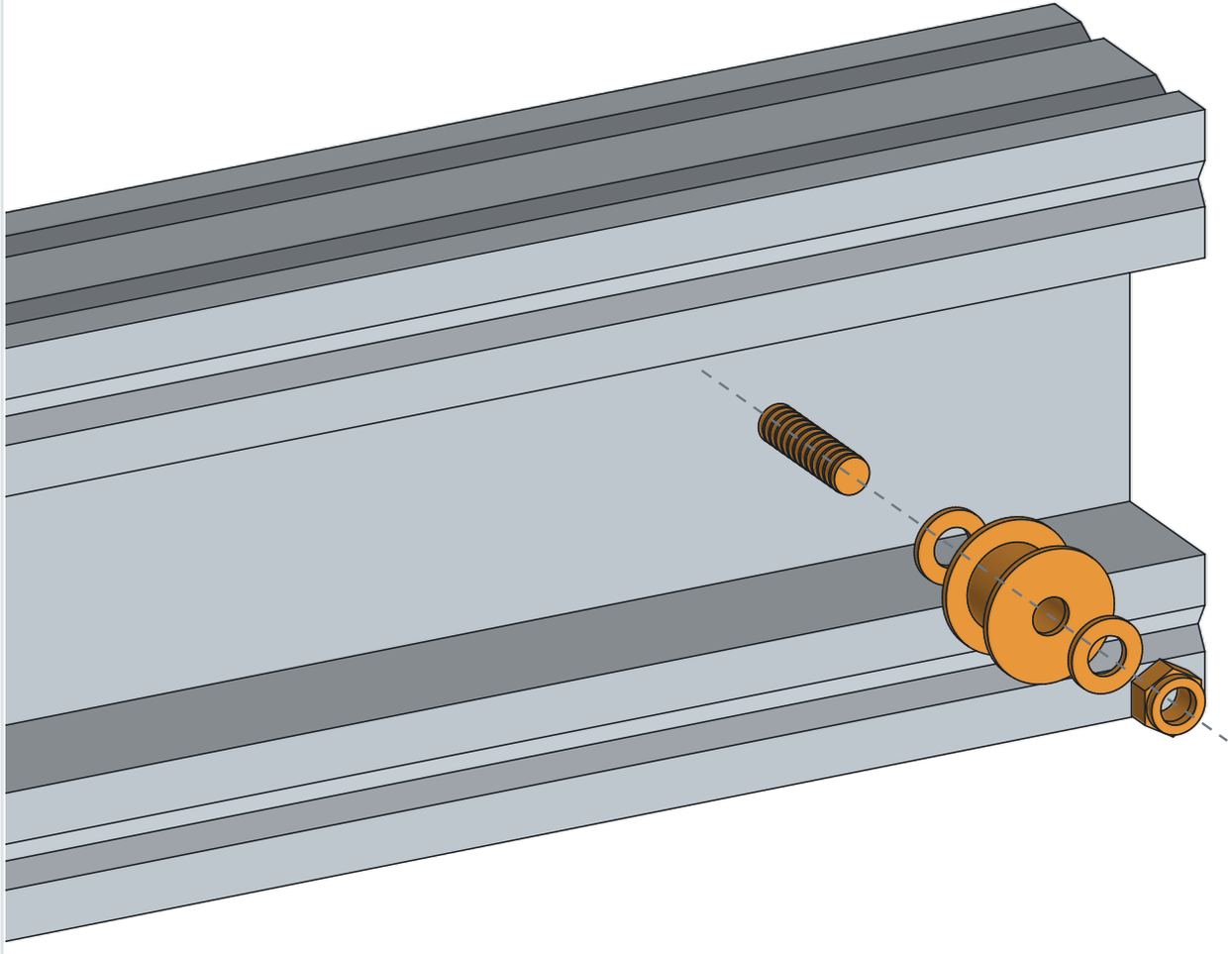
Four M5 × 10 into T-nuts in the front-lip grooves. Pulley into the channel.



Flush with the rail's end. The plate sits on the two front lips and its four screws drop into the lip grooves, 60 mm apart. The motor never moves: the belt does the travelling, and the carriage stops 5 mm short of this plate.

9 Idler at the far end

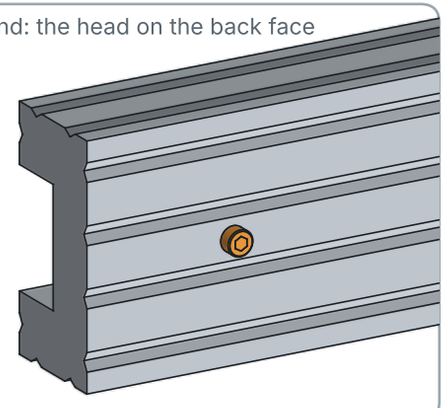
Drill 5.2 through the web, 50 from the end. M5 × 40 from the back.



- 1× M5 × 40 socket cap
- 2× M5 washer
- 1× Idler, 9 mm
- 1× M5 nyloc

5.2 drill · 4 mm

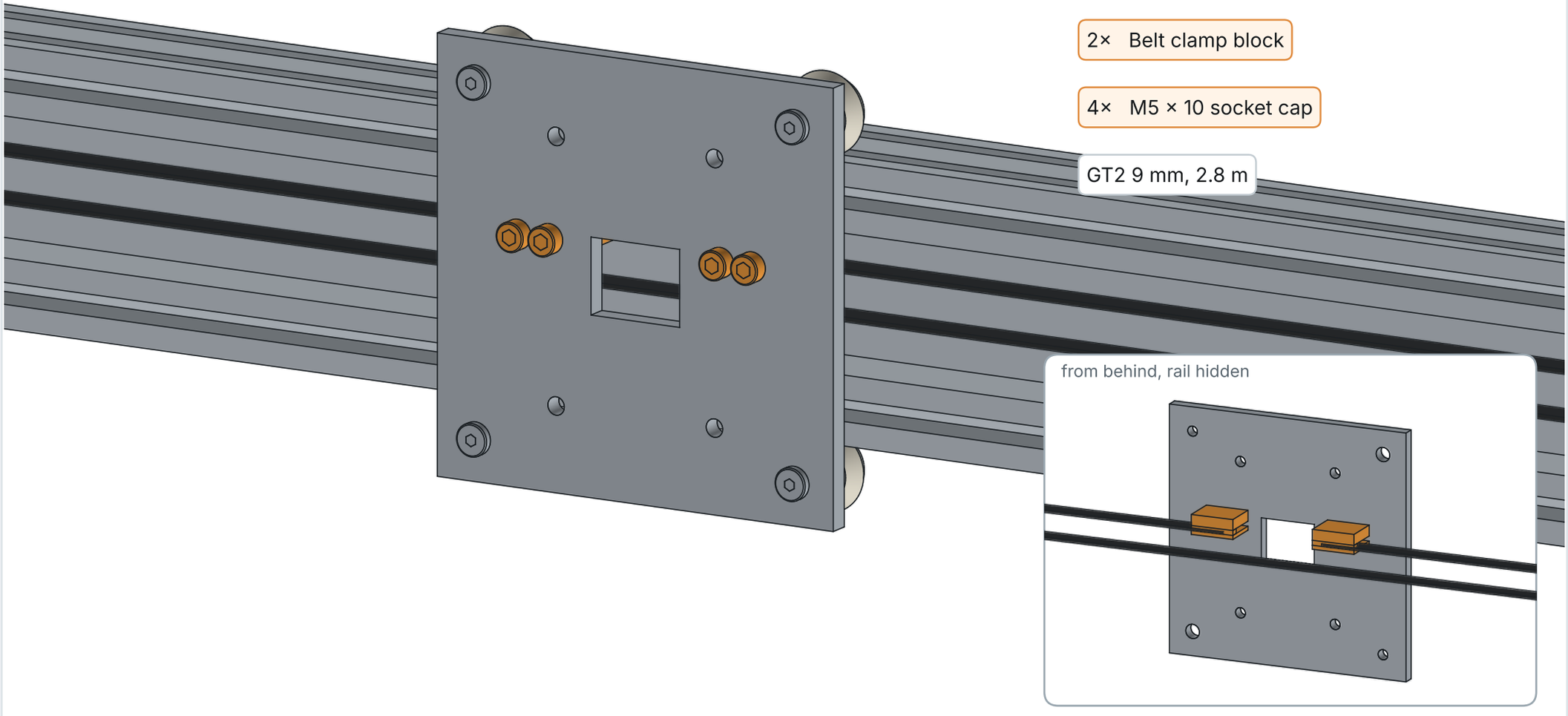
from behind: the head on the back face



Head on the back face, then inside the channel: washer, idler, washer, nyloc. Centre the hole on the web's mid-height so the idler lines up with the pulley at the other end. The nyloc snug, not tight: the idler has to spin freely.

10 Belt and clamps

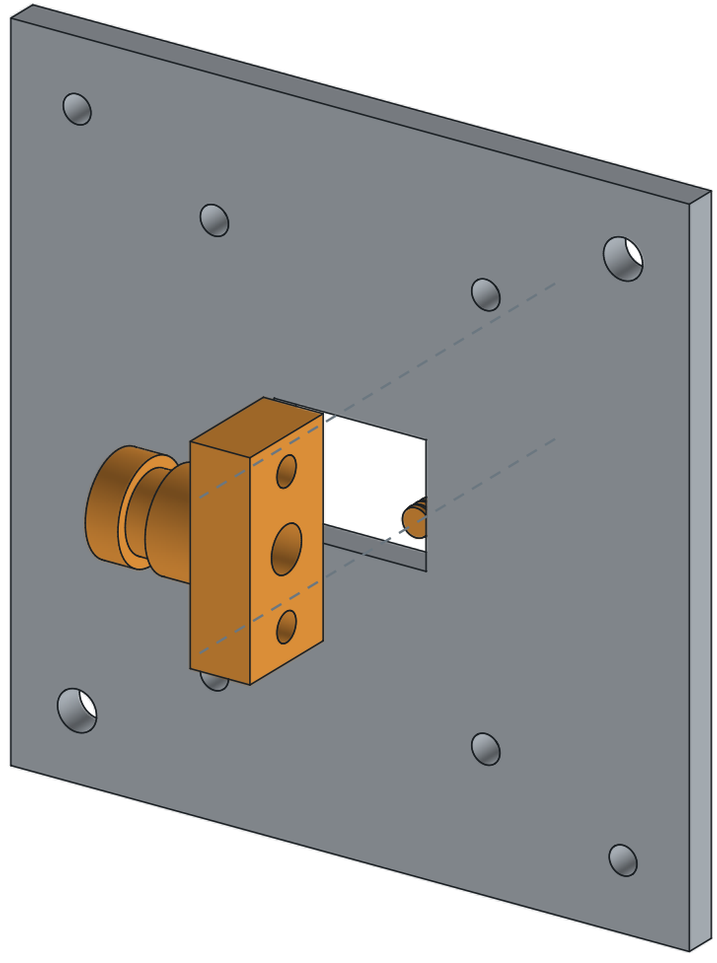
Teeth in, width along the pulley axis. Both ends clamp to the carriage.



Clamp one end. Run the belt left inside the channel, round the pulley, back along the bottom of the channel, round the idler, and into the second clamp. Pull it through to tension and lock. Plucked it gives a low clear note, not a thud. Over-tension bends the motor shaft.

11 Nut block onto the Z carriage

On the back of the plate, two M5 × 12 from the front.

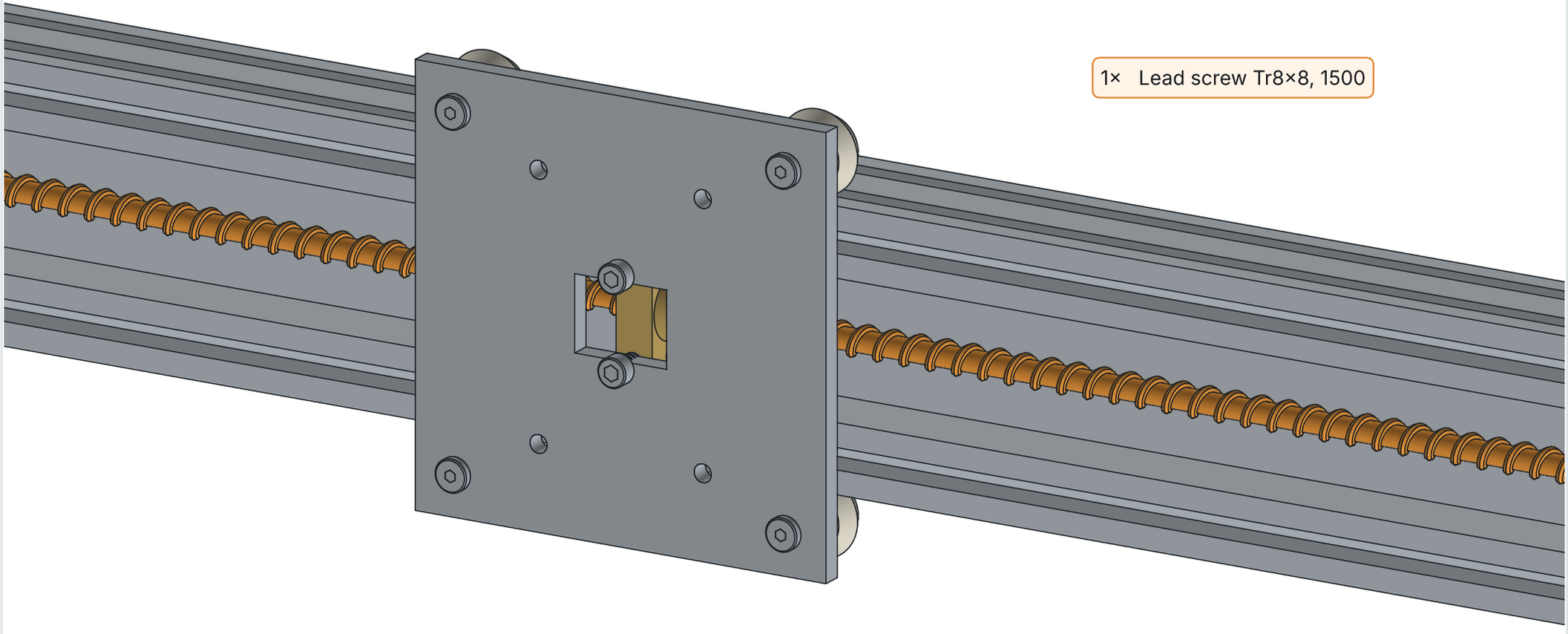


- 1× Anti-backlash nut block
- 2× M5 × 12 socket cap

Bore along the mast. The spring between the block's two halves is what takes the backlash out; do not compress it flat or the screw will bind and you will blame the motor.

12 Lead screw through the nut

Thread it in by hand, through the carriage, along the channel.

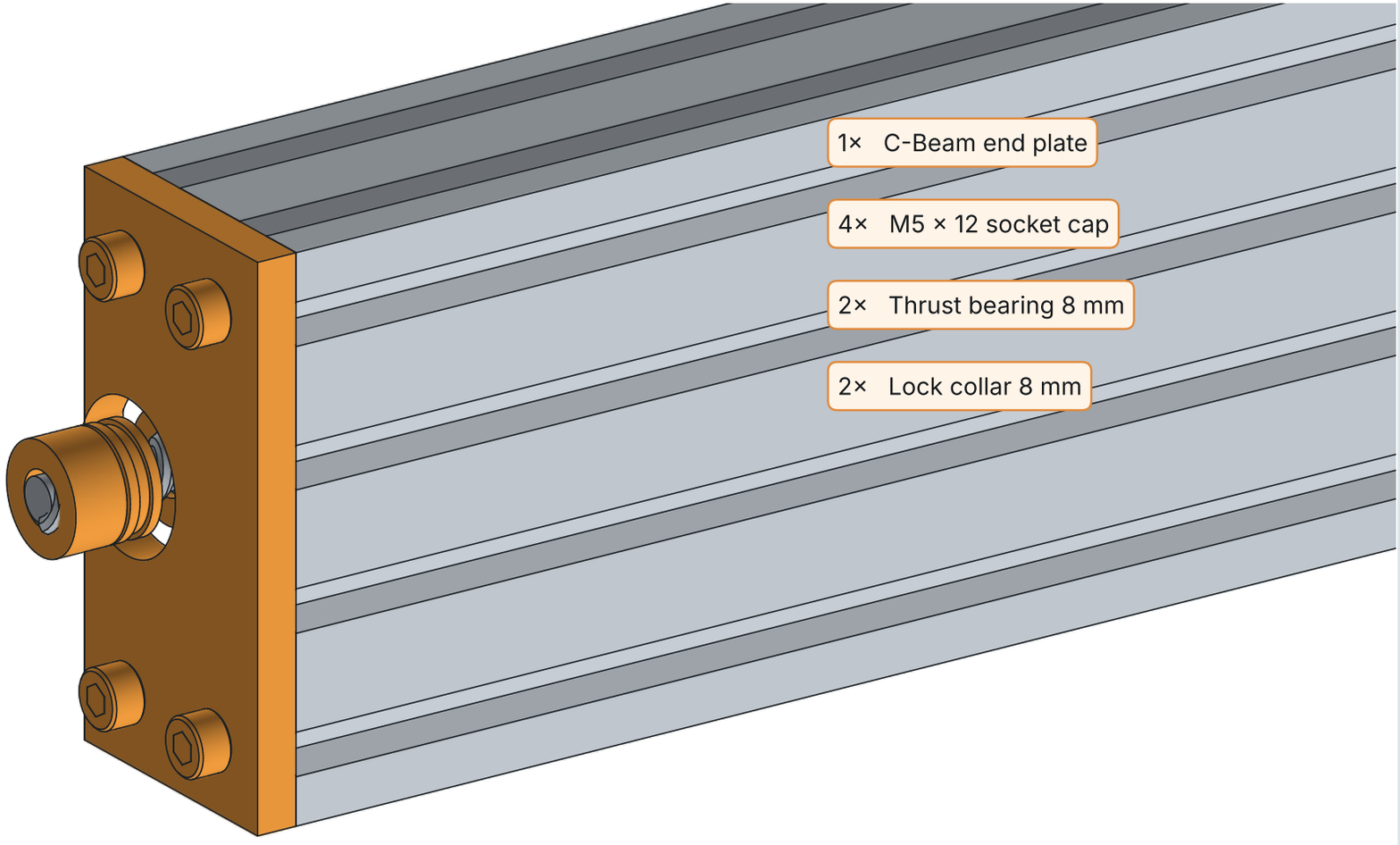


1× Lead screw Tr8×8, 1500

It sits on the channel centreline, 10 mm behind the mouth. Turn it through the nut by hand until it runs the full length with even drag. A tight spot now is a bent screw or a nut block out of line, and both are cheaper to find before the motor is on.

13 Top end · the screw hangs here

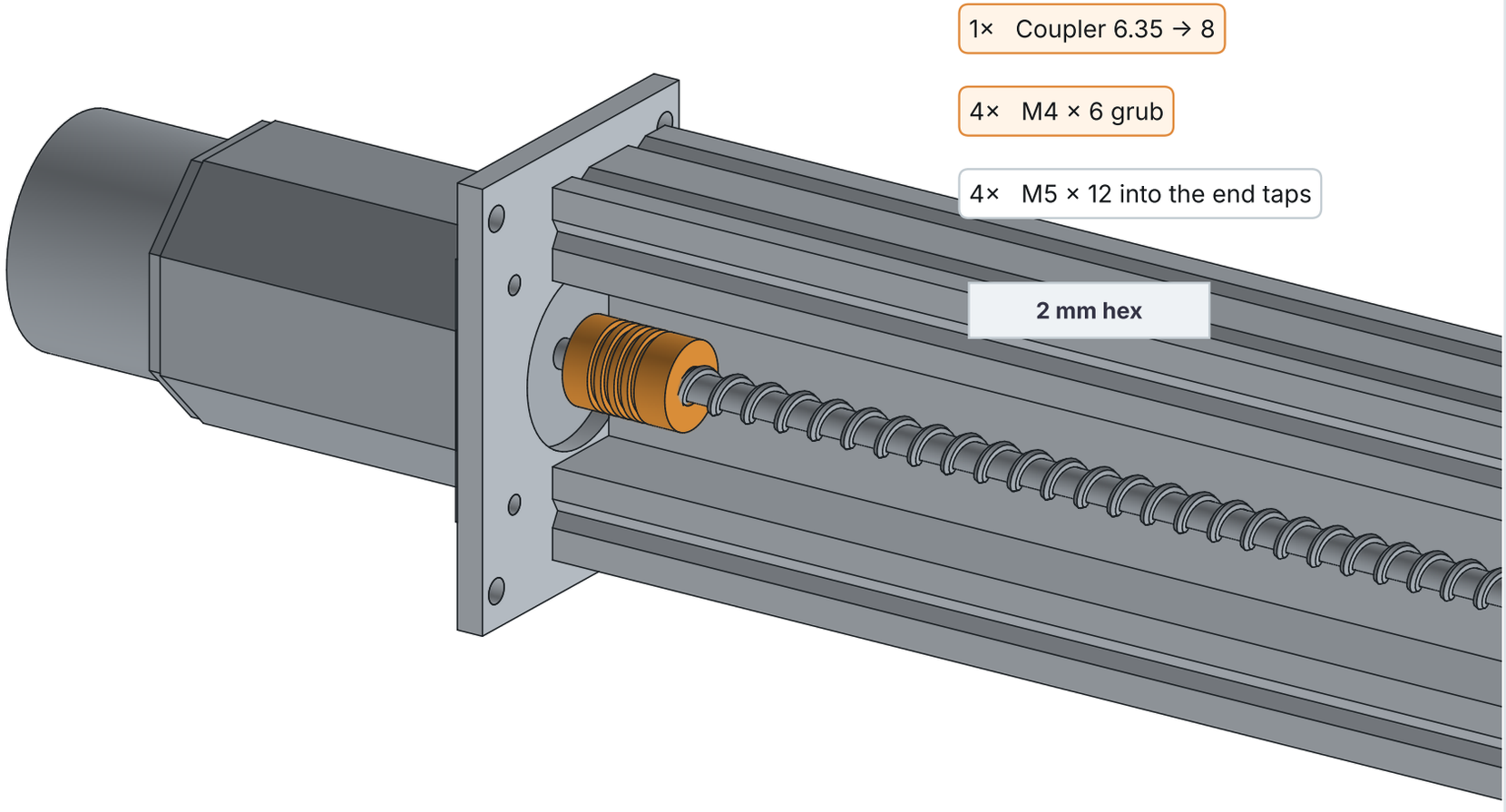
End plate, a thrust bearing each side, a lock collar each side.



The whole hanging load is AXIAL. Two thrust bearings clamp the plate between them and the collars lock them to the screw, so it hangs from the top and the coupler at the foot only ever carries torque. The end plate and collars are additions to BOM Rev A.

14 Foot · braked motor and coupler

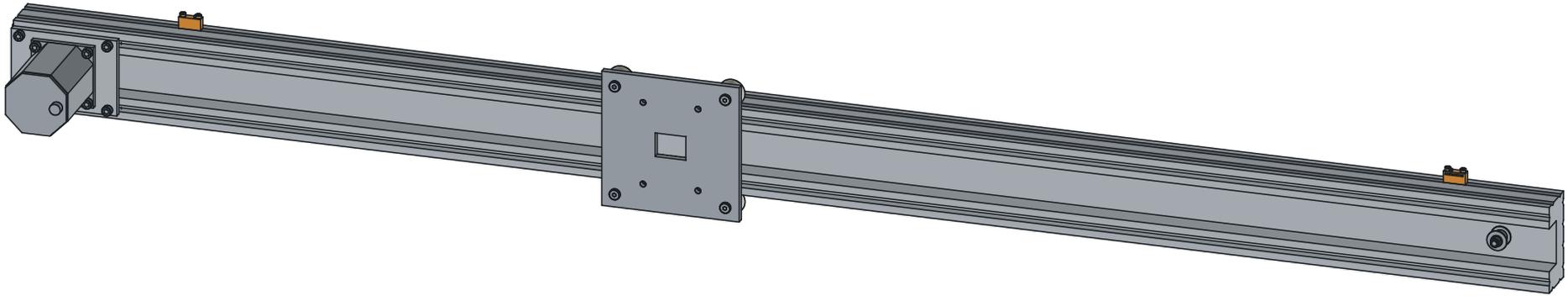
Motor plate across the end, motor outside, coupler inside.

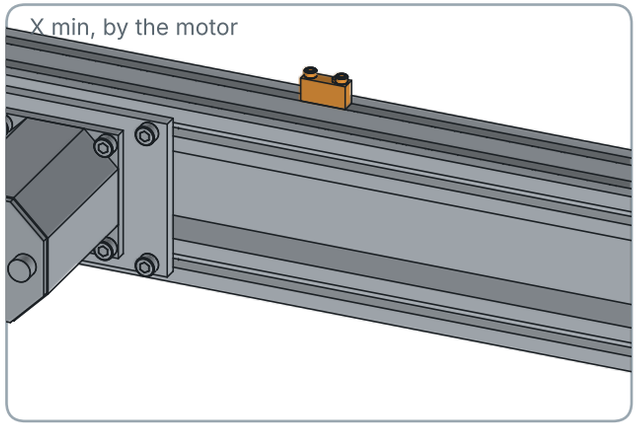


Leave 2 mm between the shaft ends inside the coupler, so it can flex. Test the brake BEFORE coupling: 24 V releases it, no power grips it. If it does not grip you find out when the power blinks and the arm lands on a tray.

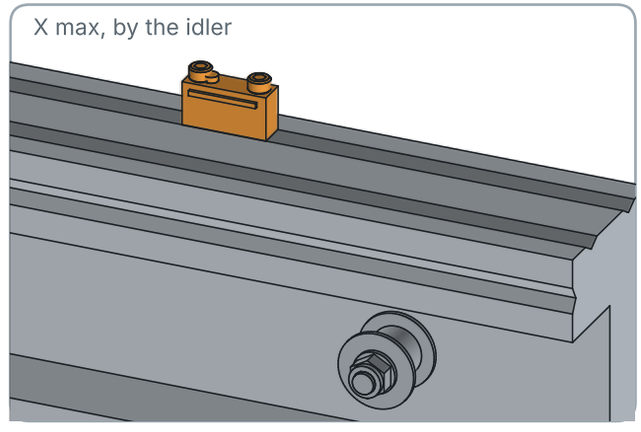
15 Limit switches · all four

X min and X max on the rail; Z min and Z max the same way on the mast.





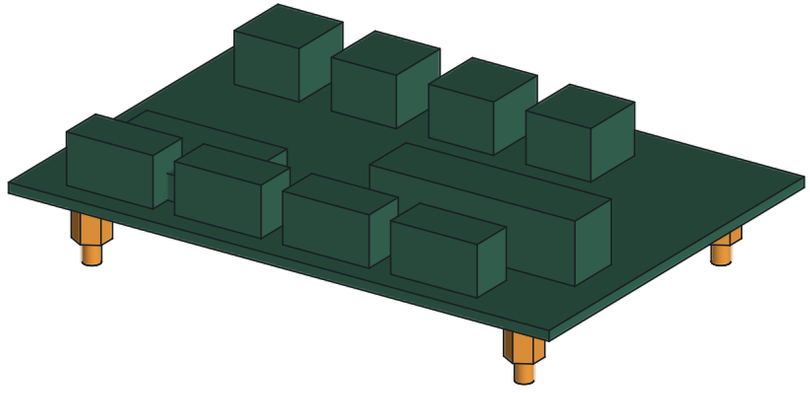
8× M3 × 10 + T-nut



In the rail's back top groove: the front one is where the wheels run. Each trips 5 to 10 mm BEFORE the hard stop. Wire them NORMALLY CLOSED, in series per axis: a cut wire then reads as a trip instead of silently disabling the limit.

16 Controller on standoffs

Four M3 standoffs. Keep the board's ground off the frame.



- 4× M3 standoff
- 4× M3 × 10 socket cap

24 V DC in
from the fuse block, via safety relay 13/14

X motor
4 wires, continuous flex, through the chain

Z motor
4 wires, plus the separate 24 V brake pair

Limits × 4
normally closed, in series per axis

USB / wifi
to the Jetson, for G-code

Standoffs, not screws straight into the extrusion: the board's ground plane must not touch the frame, or you make a loop with the motor shields.

17 First motion, before the arm

Jog, calibrate steps, home. Nothing heavy on the gantry yet.

1. **Set driver current**
   from the motor datasheet. Too high cooks the TMC2209s, too low loses steps.

2. **Jog 5 mm**
   confirm direction on both axes before anything longer.

3. **Command 100 mm**
   measure it. Correct steps/mm until commanded equals actual.

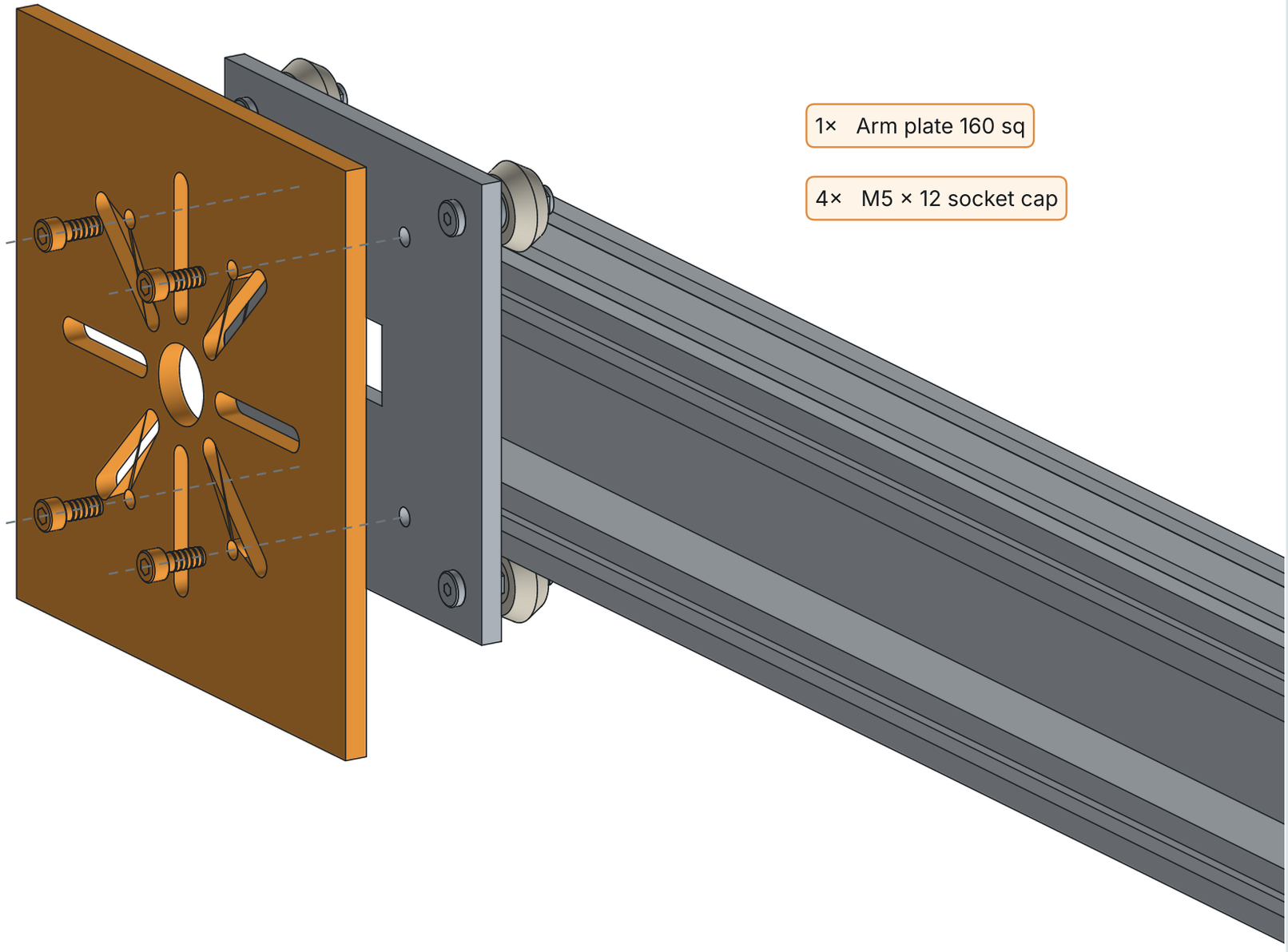
4. **Home both axes**
   and home on every power-up. FluidNC does not know where it is otherwise.

5. **Run full travel**
   ten times, both axes. Listen for a tight spot.

If the gantry is going to bind, lose steps or trip a limit wrongly, it does it now, when nothing expensive is bolted to it. That is the point of doing this before step 18.

18 Arm plate onto the Z carriage

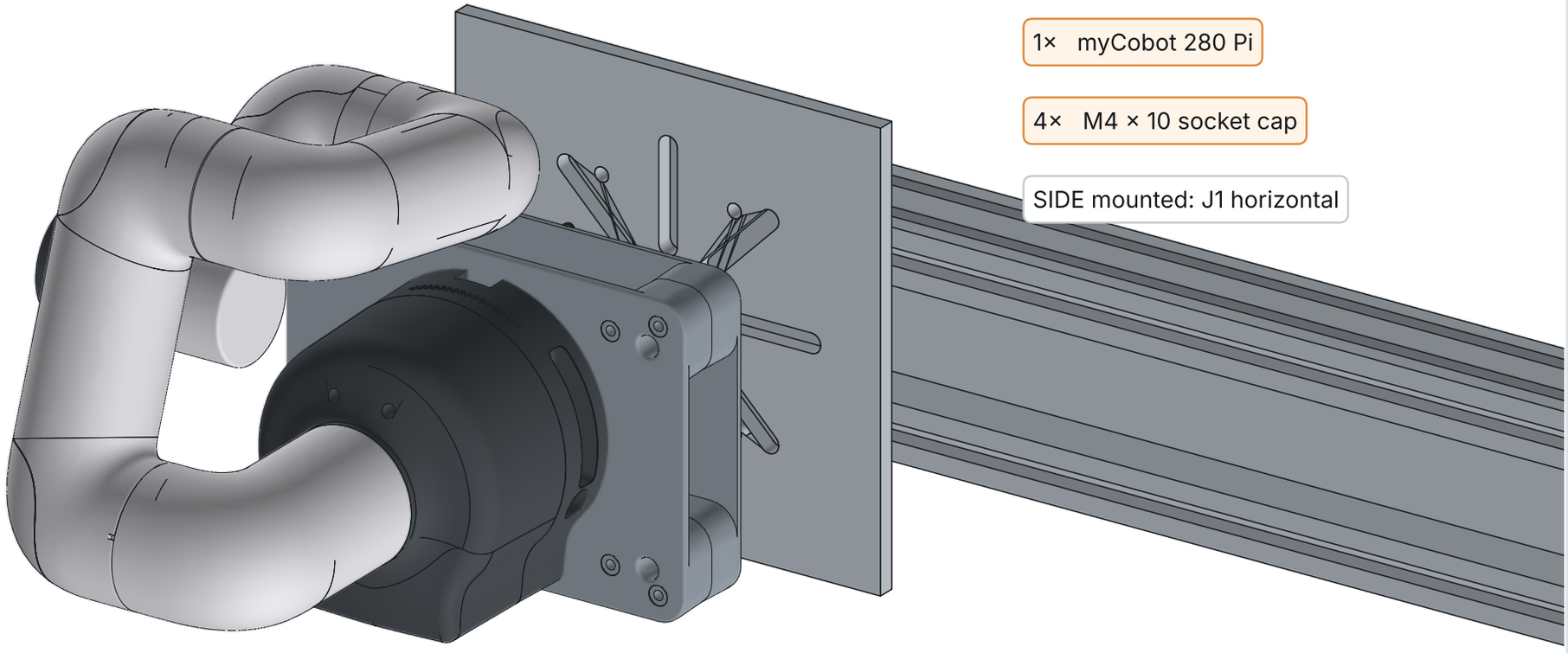
Four M5 × 12. The slots give 110 mm of adjustment.



Set the plate central in its slots for now. You will want the travel later, when hand-eye calibration says the arm wants to sit a few millimetres off where the drawing put it.

19 The arm, last

Four M4 × 10 into the base. Its own power brick.



Power the arm from its OWN brick, never from the 24 V bus. Confirm it answers over TCP before any coordinated motion. Folded as tightly as it goes without hitting itself it is 162.6 mm long, and that number decides the cabinet. See step 22.

20 Before you trust it

The safety circuit, and the checks that catch the usual faults.

- [ ] **E-stop wired, both channels**
  Two M22-K01 blocks into the G9SE-201. It cross-monitors them; one is not enough.

- [ ] **13/14 feeds motion only**
  24 V to the controller and the Z brake. Sensing and lighting stay alive.

- [ ] **E-stop mid-move**
  Motion stops, brake grips, relay refuses to reset until released.

- [ ] **Power-cut test**
  Pull the plug mid-travel. The mast must hold, not fall.

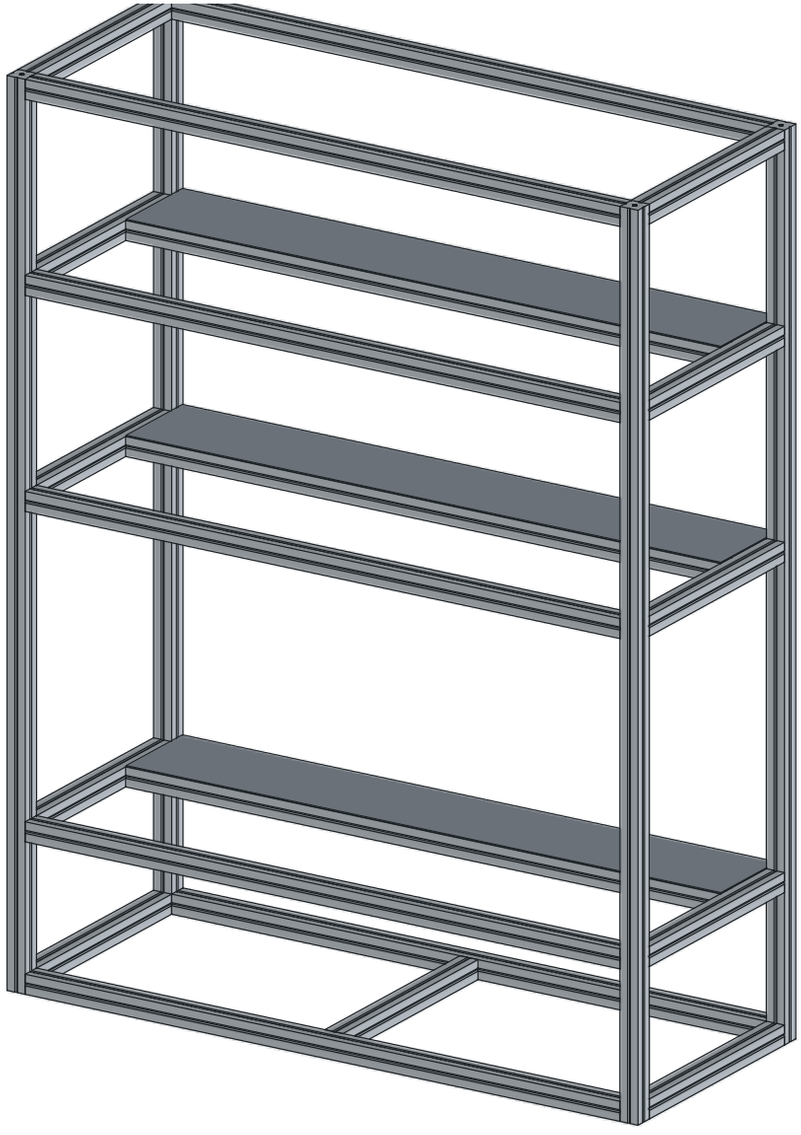
- [ ] **Limit test**
  Trip each switch by hand with the axis moving toward it.

- [ ] **Backlash check**
  Jog Z up, then down, to the same commanded point. Measure the difference.

Bench-testing the gantry without the safety relay is fine. Running the ARM without it is not.

21 The cabinet the gantry lands on

1397 × 559 × 1829, 40-series V-slot, five rail levels.



4 × 1829
uprights

13 × 1317
width rails: 5 front, 5 back, 3 under the tray's rear edge

11 × 479
depth rails (were 377 at 457 deep)

3 × 123
short rails under the tray laps

Tray line
y = 203 from the front. NOTHING of the frame behind it.

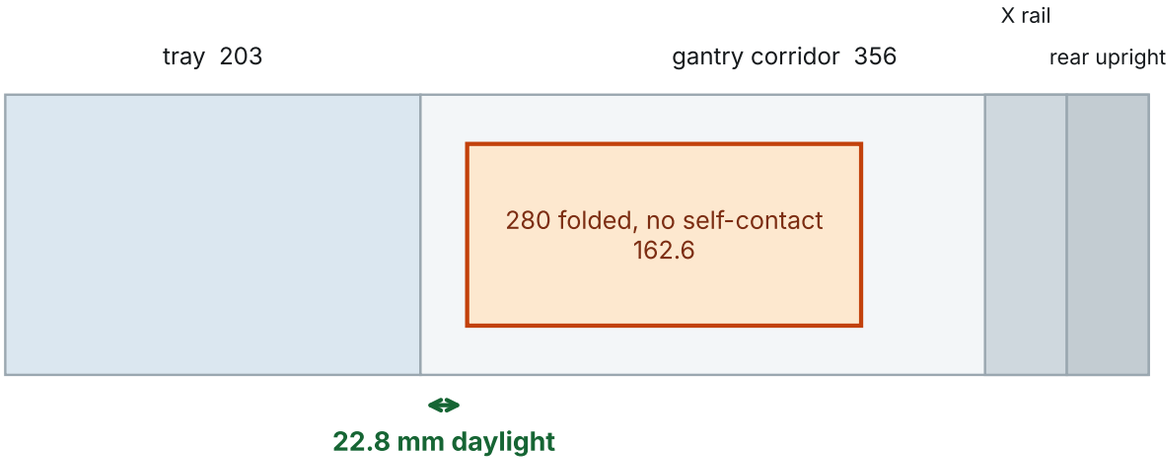
Corridor
356 mm, y = 203 to 559, all of it the gantry's

30.07 m of 4040, still 17 sticks of 2000 mm: 11 × [1317 + 479], 3 × [1829 + 123], 2 × [1317], 1 × [1829]. Only the eleven depth rails changed. The growing trays did not.

22 Why the cabinet got deeper

The one clearance that cannot be shimmed out later.

PLAN SECTION · front of the cabinet at the left · mm

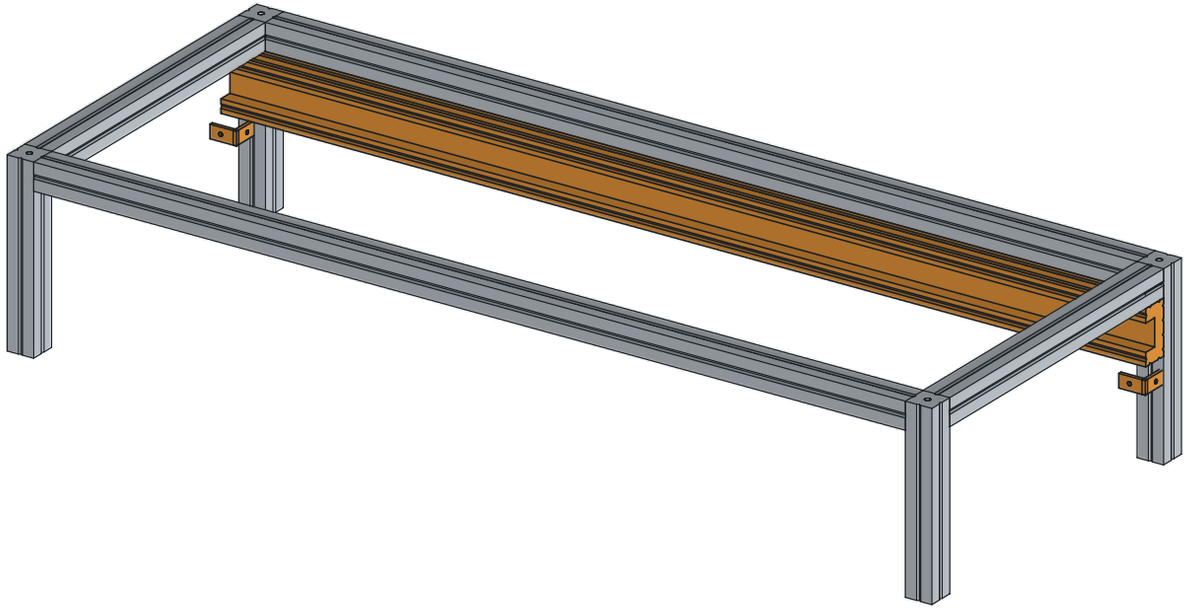


Cabinet depth	457 (Rev A)	533	559 (Rev D)
Arm J1 axis, from the front	286.4	362.4	388.4
Folded arm clears the tray by	−79.2	−3.2	22.8
Flange reaches forward to	−6.2	69.8	95.8
Gripper reaches forward to	−96.2	−20.2	5.8

The arm crosses a tray plane every time it changes tier, and the shortest fold it can reach without one link touching another is 162.6 mm, set by its own shoulder. At 457 deep it would overlap the tray by 79 mm. Measured on Elephant Robotics' mesh, collision checked.

23 X rail onto the rear uprights

Cut to 1397. Four 90° brackets. This one must be level.



seen from BEHIND the cabinet

- 1× C-Beam, cut to 1397
- 4× 90° corner bracket
- 8× M5 × 10 + T-nut



where this is

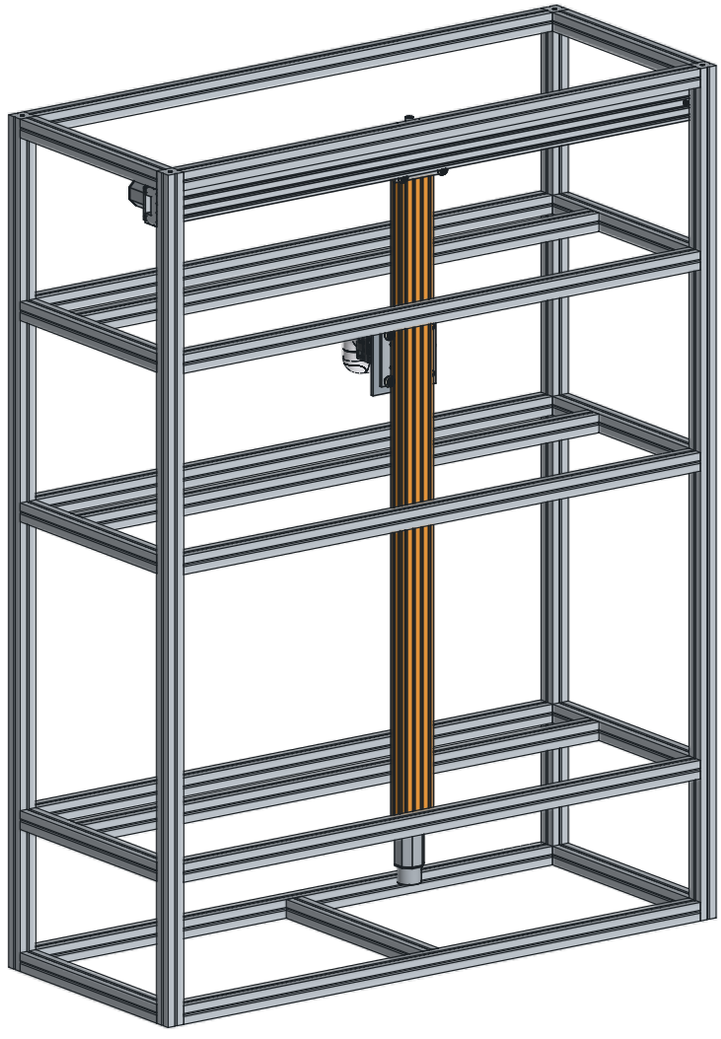
Against the FRONT faces of the rear uprights, channel facing forward, bottom edge at 1709.

Brackets on the rail's top and bottom faces at each end, beyond the carriage's travel.

Level it with a spirit level on the top face: tilt here becomes a height error across the 1126 mm of travel.

24 Hang the Z mast

X carriage and X drive on the rail first, then the mast onto the carriage.

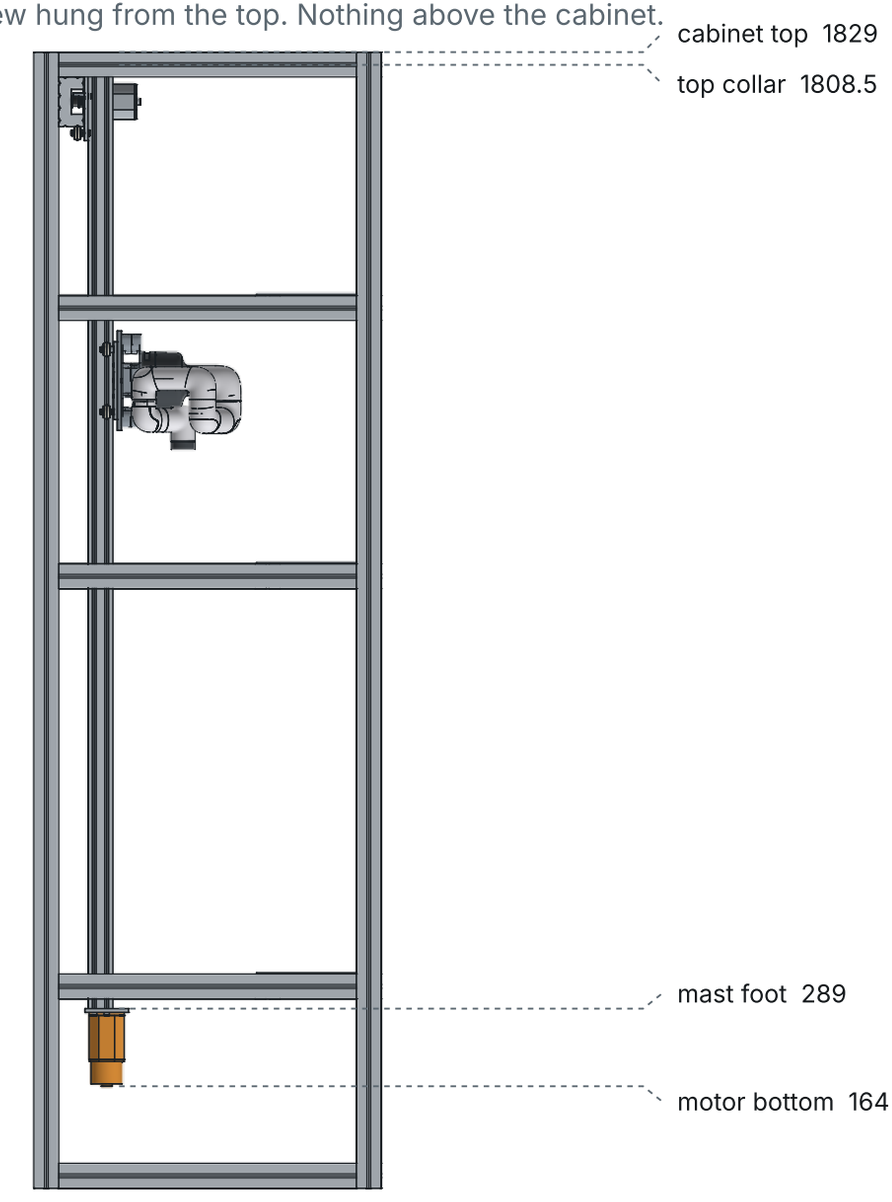


- 1× C-Beam mast, 1500, Z drive built
- 6× M5 × 10 + T-nut, mast back to plate

Six bolts, not four: they are all that stops the mast pitching forward, and the moment arm is 1.5 m. Top at 1789, foot at 289. Check plumb both ways with the carriage mid-rail, then at each end. If plumb changes across the travel, the rail is not level.

25 The Z drive, installed

Motor at the foot, screw hung from the top. Nothing above the cabinet.



1× Braked NEMA 23 + coupler

Seen from the right-hand side, front of the cabinet to the right. The top collar finishes 20.5 mm under the cabinet's top: a motor up there would stand proud of it. The screw hangs from the top bearings, so the coupler at the foot only ever carries torque.

26 Z carriage, arm plate, arm

Fold the arm before the first Z move, and keep it folded between tiers.



- 1× Z carriage
- 1× Arm plate, 160 sq × 6.35
- 1× myCobot 280 Pi, side mounted

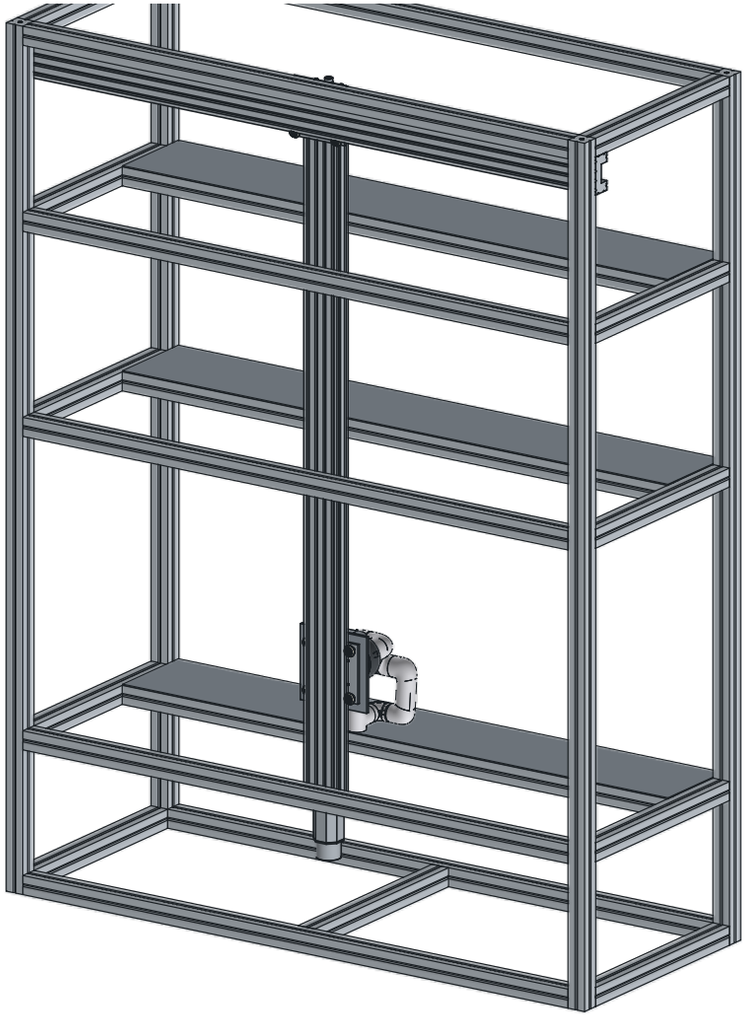
where this is

mast and frame shown cut

J1 horizontal, pointing into the tier: upright, the base radius would eat 80 mm of reach and the arm would stand 437 mm tall in a 341 mm opening. Fold: J2 −110, J3 −80, J4 −140, J5 −90, collision checked. An extended arm crossing a tray plane is the one crash this machine can do to itself at full torque.

27 Commissioned

Working envelope, and the six checks before you walk away.



X travel
1126, motor plate to idler

Z travel
1350 usable; tiers at 345, 1005, 1437

Flange reach
292.6 from J1, forward to y = 96

With the gripper
forward to y = 6: the whole tray

Folded transit
162.6 long, 22.8 clear of the tray line

Working height
263, inside the 341 tier opening

- [ ] Home X and Z, power-cycle, home again. Same zero both times.
- [ ] Fold the arm, run Z full travel. Nothing touches anything.
- [ ] E-stop mid-move: motion stops, brake grips, no reset until released.
- [ ] Pull the plug mid-travel. The mast holds.
- [ ] Jog to each tier, check the arm reaches the back row of pots.
- [ ] Hand-eye calibrate AT the tier it works at, not once at the top.

The mast is a 1.5 m cantilever off four V wheels. Expect 1 to 2 mm of forward lean at the foot that is not there at the top, which is why calibration is per tier.